Using Lasso Regression and Ego Networks to Model Synergy in Clash Royale

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1. Background

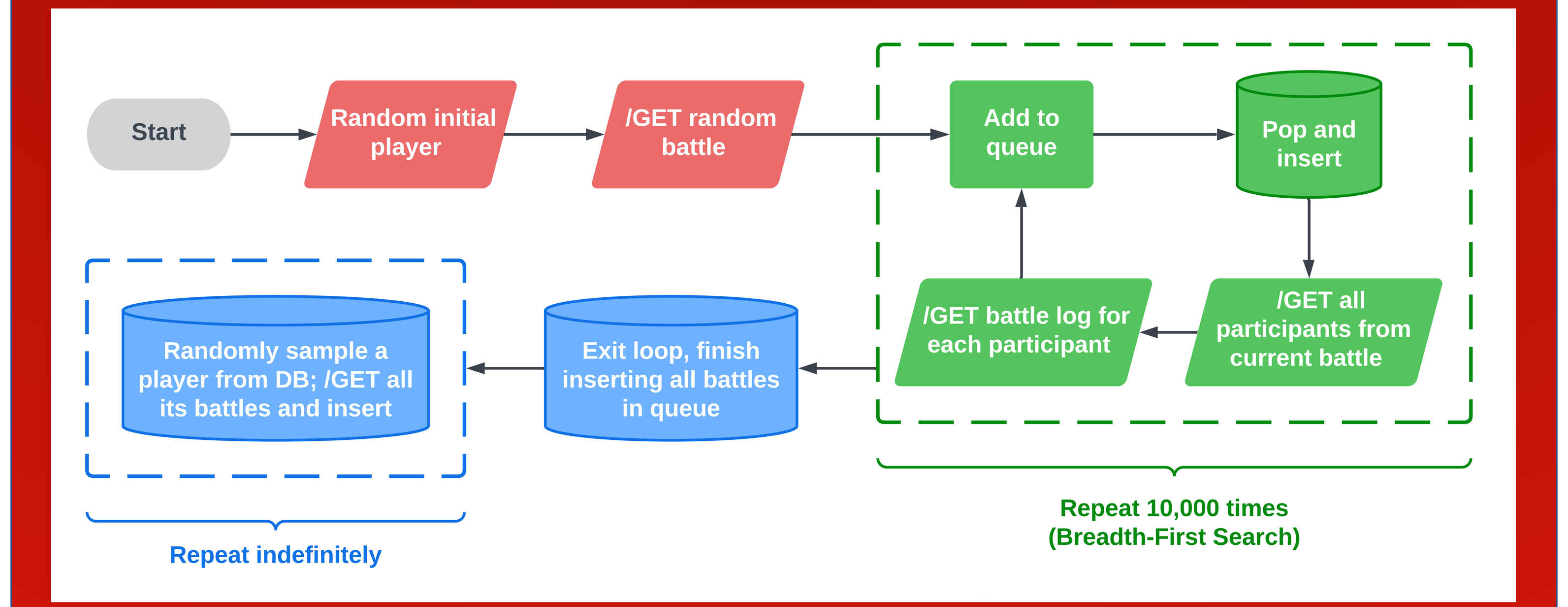
Clash Royale is a popular real-time strategy video game that was released in 2016. A typical match consists of two players battling against each other using their pre-constructed decks of eight cards. In total, there are over 100 cards available in the game.

2. Research Goal

To investigate synergistic relationships among these cards by analyzing match data.

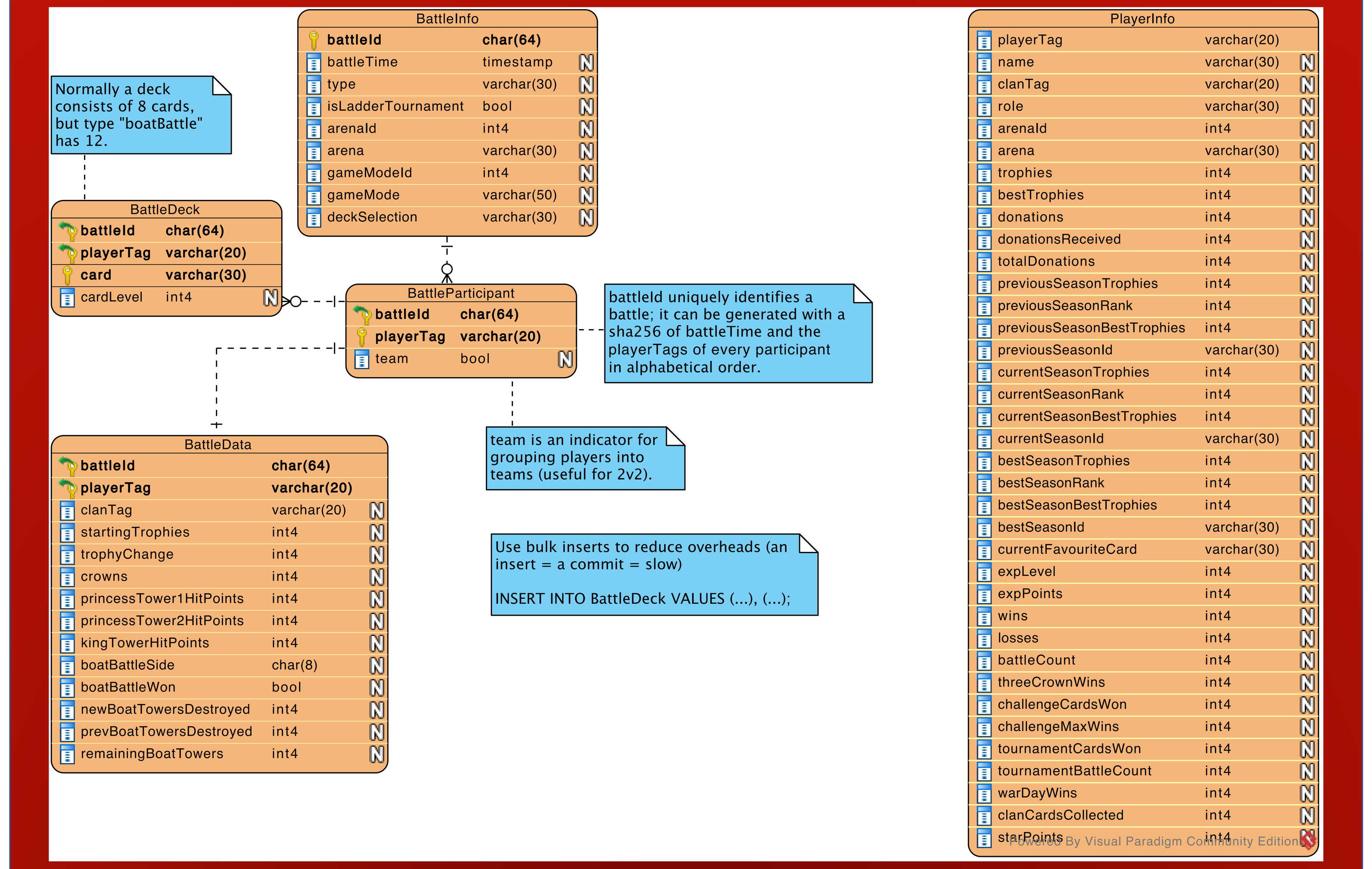
3. Data Collection

- The official Clash Royale API outputs recent match data for any player upon request
- Grow our dataset by repeatedly sending API requests
- Python is used to automate this process; the flowchart below illustrates the logic



4. Data Warehousing

- A PostgreSQL database is implemented to store data
- The Entity-Relationship diagram below maps out the structure of this database



5. Lasso Regression

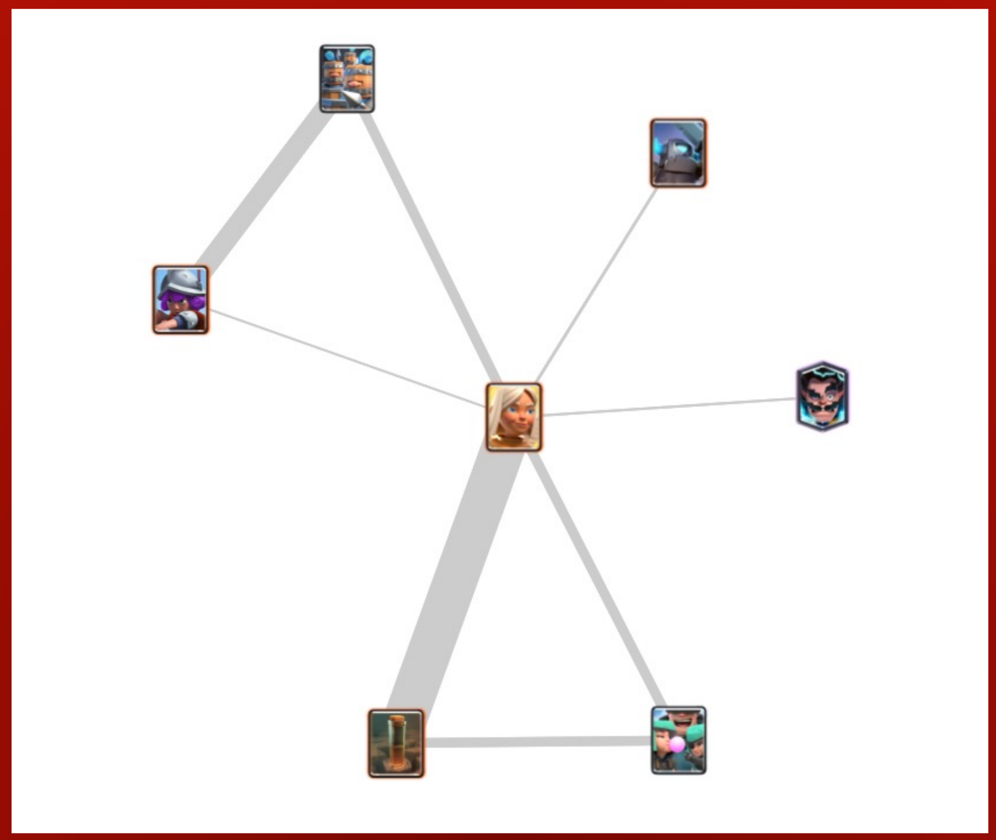
- 100 cards means 10,000 pairwise interactions
- Including all in a model is computationally inefficient
- Solution: the lasso regression algorithm effectively trims down these interactions by zeroing out insignificant coefficients
- To further reduce interaction terms, we can fit conditional models:
  - For each card, subset data with only matches using this card
  - Fit lasso regression model:

    $$matchOutcome \sim all\ other\ cards$$
  - Identify positive coefficients as synergy
- Finally, aggregate interaction terms from conditional models to form a joint model:

  $$matchOutcome \sim cards + interactions$$
- This is done with R

6. Ego Networks

- Positive interaction term coefficients from joint model indicate synergy
- Visualize these with ego networks
  - The center card is the ego
  - A network is formed with its synergies
  - Edge width corresponds to synergy strength
- This is built with the JavaScript library d3.js



7. Web App

- This is the final product
- It showcases card synergies in the form of interactive networks and heatmaps
- The app built with HTML, CSS, and JavaScript
- It is deployed using a Shiny app server

https://tomzhang255.shinyapps.io/clash_synergy/

QR code